



DIGITAL PUBLIC SERVICE PLATFORM

Super App & Mini App Platform

A unified, enterprise-grade mobile container and partner integration gateway powering secure digital public services.

Non-Bank Financial Services Authority (FSA)



AGENDA

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CamDigiKey SSO, HR Enterprise Mini App, SA Capability Generator



STRATEGIC VISION

The Challenge & The Solution

Transforming fragmented citizen applications into a unified digital public service ecosystem.

⚠️ The Fragmented Dilemma

- ✗ Citizens forced to download dozens of separate standalone government and financial applications.
- ✗ Inconsistent authentication, variable security postures, and fragmented citizen profiles across services.
- ✗ Slow partner onboarding with high engineering costs, duplicated infrastructure, and tedious release reviews.
- ✗ Heavy maintenance burden for government authorities with unstandardized mobile deployment pipelines.

✨ The Unified Super App Solution

- ✓ One App for All Services: High-performance Flutter container hosting dynamic on-demand Mini Apps.
- ✓ Automated Security Gates: Strict network isolation, domain verification, and 8-stage SAST/DAST compliance.
- ✓ Self-Service Backoffice: Instant registration wizard, live reachability checks, and version governance.
- ✓ Zero Mobile Git Mutation: Codegen and Nexus artifact hosting ensure sovereign, clean continuous deployment.



ARCHITECTURE

4 Flexible Integration Tiers

Supporting any partner technology stack with zero manual host application modification.

WebView Mini Apps (H5 / Web)

INSTANT WEB DEPLOY

Interactive responsive web applications running in a sandboxed Web container with bidirectional SuperAppJSBridge and native hardware capability access (camera, location, storage).

Flutter Package (Git / Nexus)

HIGH PERFORMANCE

High-performance in-app micro-apps integrated via Git repository source OR hosted Dart Pub artifacts from Sonatype Nexus CE with offline caching and native 120 FPS transitions.

Native SDK Frameworks (.aar / .xcframework)

ENTERPRISE BINARY

Platform-specific binary modules integrated automatically via Native SDK Codegen and Universal Native Launcher without manual mobile codebase edits.

Deep Link & App Links (OS Scheme)

EXTERNAL ECOSYSTEM

Seamless OS-level URL scheme delegation (e.g. cambodia-tax://) to external standalone applications with automated App Store / Play Store fallback routing.



DEVELOPER EXPERIENCE

Partner Self-Service & Developer Portal

Next.js Backoffice Admin Portal with live URL reachability, capability proposals, and regulatory vetting.

6-Step Onboarding Wizard

Guided step-by-step registration capturing app metadata, identity branding logos, technical integration endpoints, and ownership credentials.

Live URL Reachability Probes

Real-time debounced HTTP & TCP probes verifying production URLs, privacy policy endpoints, and store fallbacks during data entry.

Flexible Policy Disclosures

Supports live HTTPS regulatory URLs or self-contained offline Markdown policy documents with built-in compliance templates.

 **Super App**
SUPER APP GATEWAY

DASHBOARD

 Dashboard

 Notifications

ORGANIZATION

 Organizations

 Users & Roles

ECOSYSTEM

 Integration Guidelines

 Mini Apps

 Permissions

 Super App

REVIEW

 Review Queue

SYSTEM

 Audit Logs

 Settings & Profile

 **Super Admin**
superadmin@example.com

FinTech Center General Secretariat of FSA

Download APK

API: Online

SA Super Admin
superadmin@example.com

← Register Mini App

Deploy a new service to the Super App gateway

1 Basic Info

2 Team

3 Integration

4 Permissions

5 Security

6 Review

GENERAL INFORMATION

Mini App Name *

e.g. Core Banking App

Mini App ID (Auto-generated) *

miniapp_core_banking_8f32a1

Organization *

FTC - Financial Technology Center (FTC)

Logo *

Click to upload image or drag & drop... Browse

PNG, JPG, SVG or WebP (square recommended, max 5MB)

Short Description

One sentence summary

Optional

Full Description

Comprehensive details regarding the purpose and functionality...

Optional



ENTERPRISE SECURITY • STRICT

8 Automated Security Scanners & Gates

Multi-layered validation pipeline executed automatically by Jenkins CI/CD on every package submission.

01. Cryptographic SHA-256 [PASS ✓]

02. Gitleaks Secret [PASS ✓]

03. Semgrep & AST Guard (SAST) [PASS ✓]

04. Trivy & OSV Database (SCA/CVE) [PASS ✓]

05. Super App Gatekeeper [PASS ✓]

06. Syft CycloneDX (SBOM) [PASS ✓]

07. License Compliance Scanner [PASS ✓]

08. ClamAV & YARA Malware Scan [PASS ✓]

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←

native-sdk-2 TESTING

Edit Configuration

...

miniapp_native_sdk_2_5ca998 • Native SDK • FTC

Manual Sandbox Testing Phase

The test container has been compiled. Test the Mini App using the **Sandbox Preview** or **Download Test APK** on Android devices. When verified, SA Admin can Activate the app.

Launch Sandbox

General Information

Team & Support

Technical Integration

Permissions & Capabilities

Versions & Releases

Security & Compliance Report PASSED

100 SCORE

Automated Security & Compliance Report PASSED

Target: N/A

Report generated: 10/1/2026, 11:49:42 AM

Configure Checks

Re-Run Security Scan

ACTIVE SECURITY SCAN PROFILE (7 CHECKS ENABLED)

Static Application Security Testing (SAST) Semgrep / AST Guard

Dependency Vulnerability Scan Trivy / OSV / Audit

Secret & API Key Leak Detection Gitleaks / TruffleHog

Host Capability Gatekeeper Audit Super App Gatekeeper

Software Bill of Materials (SBOM) Syft / CycloneDX

Malware & Binary Signature Scan ClamAV / YARA

Open Source License Compliance FOSSA / License-Checker

Edit Checks

JENKINS AUTOMATED VALIDATION PIPELINE

Real-time execution log of the security stages orchestrated by Jenkins:

1. Ingestion & Integrity Verification

Cryptographic SHA-256 Digest

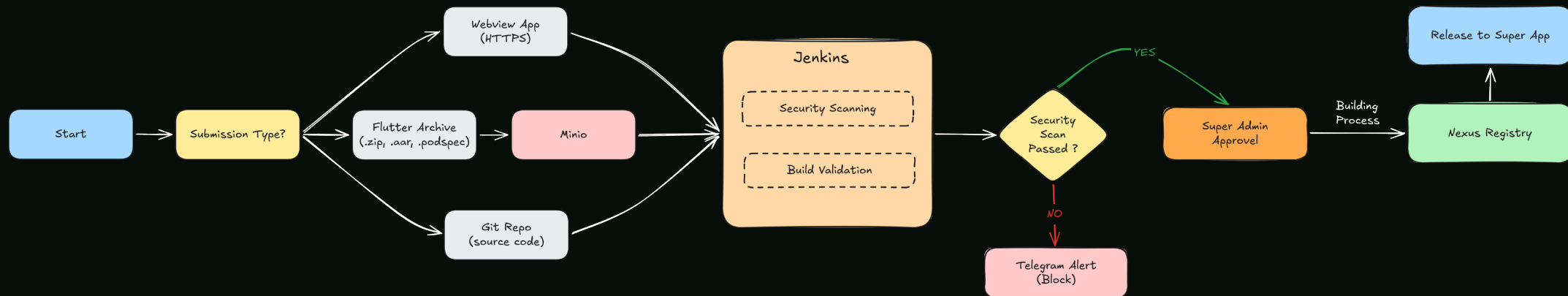
COMPLETED



AUTOMATED ASSEMBLY

Security Scan Flow

Release-mode compilation with Jenkins, Sonatype Nexus Repository CE, and automated SDK codegen.



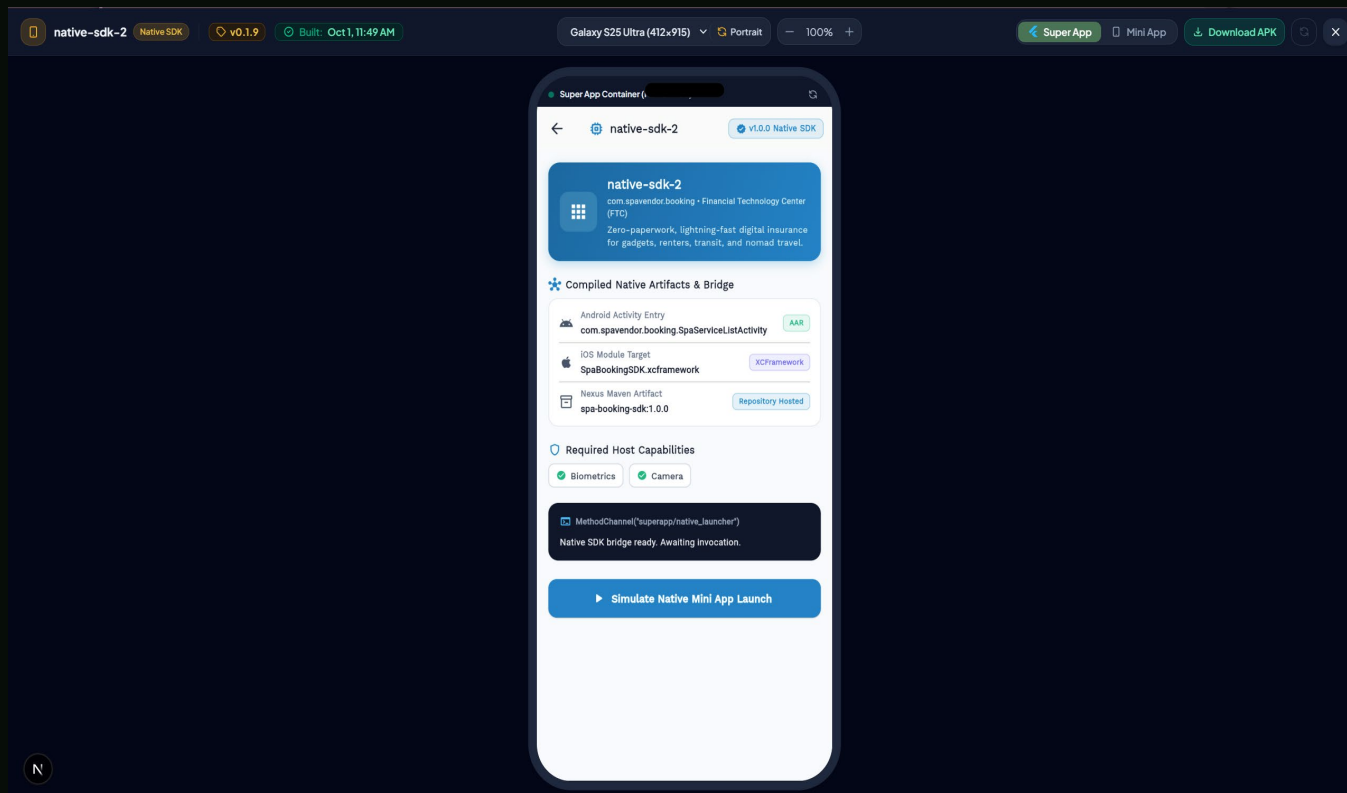


TESTING & VERIFICATION

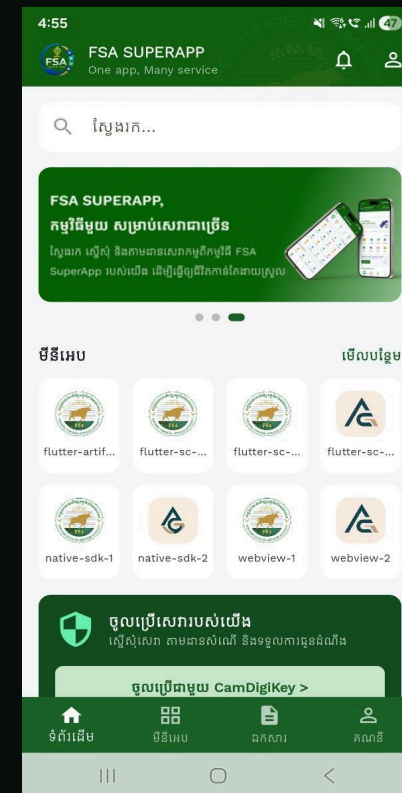
Dual-Environment Testing & QA Engine

Instant in-browser Flutter Web simulation combined with physical Android device verification.

Web:



Mobile:





STATUS & ACHIEVEMENTS

Current Milestones & Operational Status

Working prototype and governance infrastructure fully validated end-to-end.

4 Tiers Verified (90%)

WebView Mini App, Flutter Package, Native SDK, and Deep Link validated end-to-end with live sample services.

2-Stage Automated CI/CD

Jenkins compilation pipelines & Sonatype Nexus CE repository operational with automated release packaging.

Production Infrastructure

Consolidated database schema, real-time Socket.IO notifications, and high-reliability Docker containers published.

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Super App Administration

Global ecosystem metrics, pending submissions, release versions, and platform infrastructure.

+ Register Mini App

Review Pending (0)

Ecosystem Architecture

TOTAL MINI APPS

8

PENDING REVIEWS

0

SUPPORTED PERMISSIONS

5

TEST BUILD VERSION

v0.2.7

Testing

OFFICIAL RELEASE

v1.0.0

Production

Ecosystem Services & Infrastructure

Operational status of object storage, messaging bots, artifact registries, and CI/CD pipelines.

Manage Infrastructure

STORAGE ENGINE

Community / Free Edition

localhost:9000

TELEGRAM GATEWAY

Bot Gateway

@superapp_notification_bot

NEXUS REGISTRY

Sonatype Repository

http://localhost:8081

CI/CD ENGINE

Jenkins Automation

http://localhost:8085



STRATEGIC ROADMAP

Next Integrations

Upcoming high-impact integrations and enterprise ecosystem expansion.



HR Mini App



SA Capability Generator



CamDigiKey Integration



👉 Q&A / THANK YOU

Thank You!

Empowering national-scale digital public services and partner ecosystems through a unified, secure Super App platform.

Non-Bank Financial Services Authority (FSA) • Core Engineering Team